

Basal Rate Tuning History

Andrew · Type 1 · Dexcom + Loop (closed-loop) · June 2026

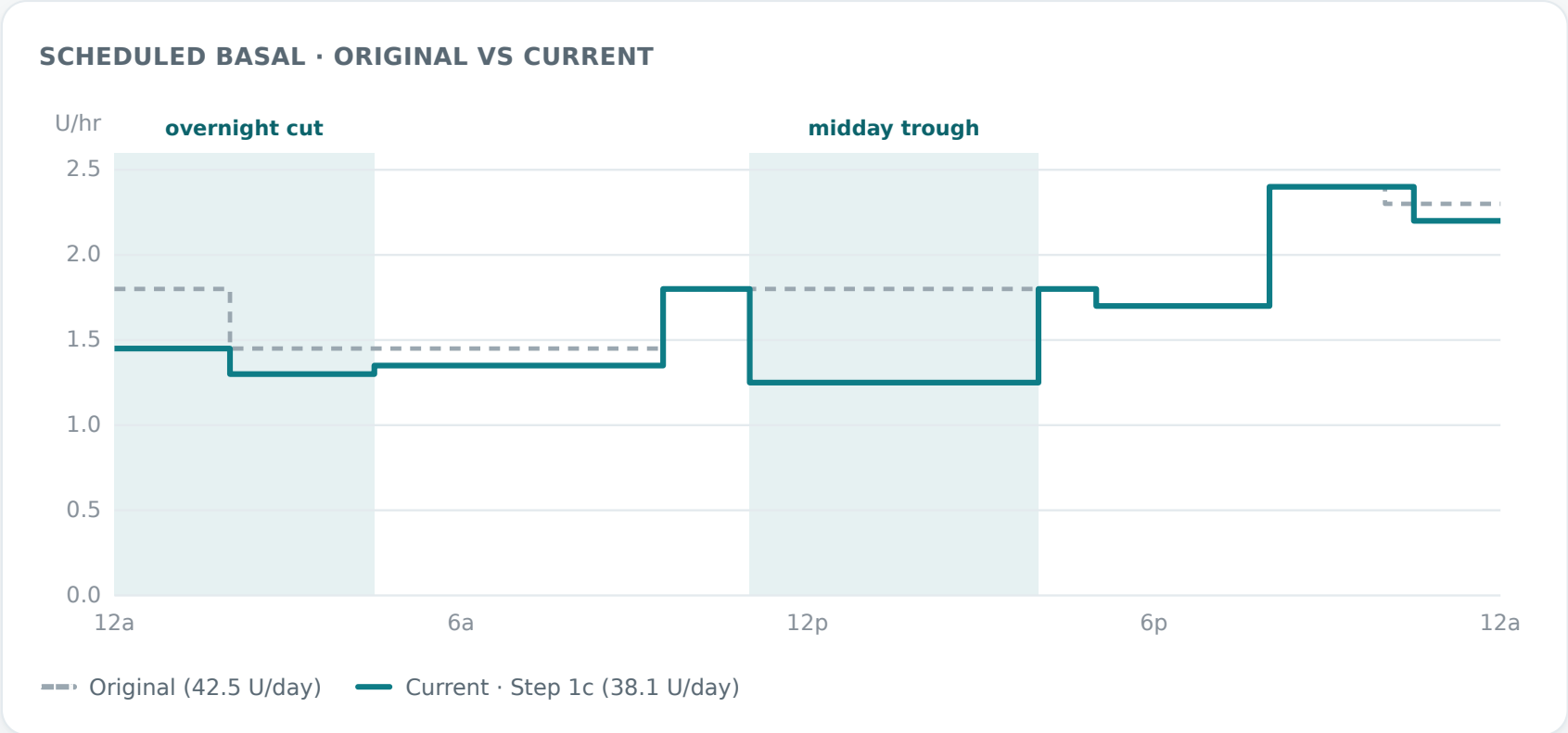
Record of scheduled-basal changes, entered in the Loop app and verified via Nightscout. **Not medical advice.**

OVERNIGHT LOWS (00-04)
12.5% → ~3%
time below 70 mg/dL

TIME IN RANGE NOW
93%
~95% excl. sensor compression

GMI · EST. A1C
5.9%
held — no loss of control

TOTAL DAILY BASAL
-4.4 U/day
-10% (42.5 → 38.1 U)



HOW WE GOT HERE · FOUR STEPS

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Baseline through Jun 12

Starting schedule (6 segments).
90-day review: excellent A1c/TIR but too many lows — overnight (00–04) 12.5% below 70, midday to 16%. Loop was already suppressing scheduled basal ~40% overnight and suspending ⅓ of the night, yet still going low → the schedule itself was too high.

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Step 1 Jun 13 37.5 U/day (-5)

Overnight cut, new midday trough (11:00–16:00 @ 1.25), late-dinner taper (22:30 → 1.8).
Bring the schedule down toward what Loop was actually delivering; carve out the midday low window.
RESULT Overshot. First night ran HIGH (mean 156) — the 22:30 evening taper was too deep and glucose escaped upward. The midday trough worked immediately.

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Step 1b Jun 13 39.05 U/day (+1.5)

Walk back the evening (22:30 → 2.20) and nudge deep-overnight up (00:00 1.45, 02:00 1.30, 04:30 1.55). Keep the midday trough.
The evening needed coverage; the deep-overnight cut was fine. Restore evening, keep overnight low.
RESULT Overnight SOLVED — %<70 12.5 → 3.1, TIR 96.9% (sensor day-1 excluded). Revealed two leftovers: dawn now ran a touch low, and midday lows that turned out to be meal/bolus-driven, not basal.

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Step 1c Jun 20 38.05 U/day (-1)

Single change: cut the dawn rate (04:30 1.55 → 1.35).
Change one variable so the next read is clean. Dawn was the clear remaining basal lever.
RESULT Dawn fixed (9.7 → 5.6% <70); held through the G6→G7 sensor switch. Current schedule — a keeper. Remaining overnight dips are G7 compression artifact, not basal.

Method. Each change was entered by the user in the Loop app (Therapy Settings → Basal Rates) and verified via the profile Nightscout received. One variable was changed at a time and reviewed ~1 week later. Rationale throughout: Loop was suppressing the scheduled rate and still going low, so the schedule was lowered toward Loop's actual delivered rate. **A further optional step (Step 2)** — trimming the daytime shoulders to ~35.9 U/day — is documented but not applied; the current schedule is holding well. Not medical advice.